

Class 18: Pertussis and the CMI-PB project

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Background

Pertussis (more commonly known as **whooping cough**) is a highly contagious respiratory disease caused by the bacterium *Bordetella pertussis* (Figure 1). People of all ages can be infected leading to violent coughing fits followed by a characteristic high-pitched “whoop” like intake of breath. Children have the highest risk for severe complications and death. Recent estimates from the WHO indicate that ~16 million cases and 200,000 infant deaths are due to pertussis annually (Black et al. 2010).

1. Investigating pertussis cases by year

Q1.

With the help of the R “addin” package `datapasta` assign the CDC pertussis case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

```
library(datapasta)
cdc <- read.csv("U.S. Reported Pertussis Cases_ 1922 - 2025.csv")
cdc
```

	Year	Number.of.Reported.Pertussis.Cases	Data.Status
1	1922	107,473	Finalized
2	1923	164,191	Finalized
3	1924	165,418	Finalized
4	1925	152,003	Finalized
5	1926	202,210	Finalized
6	1927	181,411	Finalized
7	1928	161,799	Finalized
8	1929	197,371	Finalized
9	1930	166,914	Finalized
10	1931	172,559	Finalized
11	1932	215,343	Finalized

12	1933	179,135	Finalized
13	1934	265,269	Finalized
14	1935	180,518	Finalized
15	1936	147,237	Finalized
16	1937	214,652	Finalized
17	1938	227,319	Finalized
18	1939	103,188	Finalized
19	1940	183,866	Finalized
20	1941	222,202	Finalized
21	1942	191,383	Finalized
22	1943	191,890	Finalized
23	1944	109,873	Finalized
24	1945	133,792	Finalized
25	1946	109,860	Finalized
26	1947	156,517	Finalized
27	1948	74,715	Finalized
28	1949	69,479	Finalized
29	1950	120,718	Finalized
30	1951	68,687	Finalized
31	1952	45,030	Finalized
32	1953	37,129	Finalized
33	1954	60,886	Finalized
34	1955	62,786	Finalized
35	1956	31,732	Finalized
36	1957	28,295	Finalized
37	1958	32,148	Finalized
38	1959	40,005	Finalized
39	1960	14,809	Finalized
40	1961	11,468	Finalized
41	1962	17,749	Finalized
42	1963	17,135	Finalized
43	1964	13,005	Finalized
44	1965	6,799	Finalized
45	1966	7,717	Finalized
46	1967	9,718	Finalized
47	1968	4,810	Finalized
48	1969	3,285	Finalized
49	1970	4,249	Finalized
50	1971	3,036	Finalized
51	1972	3,287	Finalized
52	1973	1,759	Finalized
53	1974	2,402	Finalized
54	1975	1,738	Finalized

55	1976	1,010	Finalized
56	1977	2,177	Finalized
57	1978	2,063	Finalized
58	1979	1,623	Finalized
59	1980	1,730	Finalized
60	1981	1,248	Finalized
61	1982	1,895	Finalized
62	1983	2,463	Finalized
63	1984	2,276	Finalized
64	1985	3,589	Finalized
65	1986	4,195	Finalized
66	1987	2,823	Finalized
67	1988	3,450	Finalized
68	1989	4,157	Finalized
69	1990	4,570	Finalized
70	1991	2,719	Finalized
71	1992	4,083	Finalized
72	1993	6,586	Finalized
73	1994	4,617	Finalized
74	1995	5,137	Finalized
75	1996	7,796	Finalized
76	1997	6,564	Finalized
77	1998	7,405	Finalized
78	1999	7,298	Finalized
79	2000	7,867	Finalized
80	2001	7,580	Finalized
81	2002	9,771	Finalized
82	2003	11,647	Finalized
83	2004	25,827	Finalized
84	2005	25,616	Finalized
85	2006	15,632	Finalized
86	2007	10,454	Finalized
87	2008	13,278	Finalized
88	2009	16,858	Finalized
89	2010	27,550	Finalized
90	2011	18,719	Finalized
91	2012	48,277	Finalized
92	2013	28,639	Finalized
93	2014	32,971	Finalized
94	2015	20,762	Finalized
95	2016	17,972	Finalized
96	2017	18,975	Finalized
97	2018	15,609	Finalized

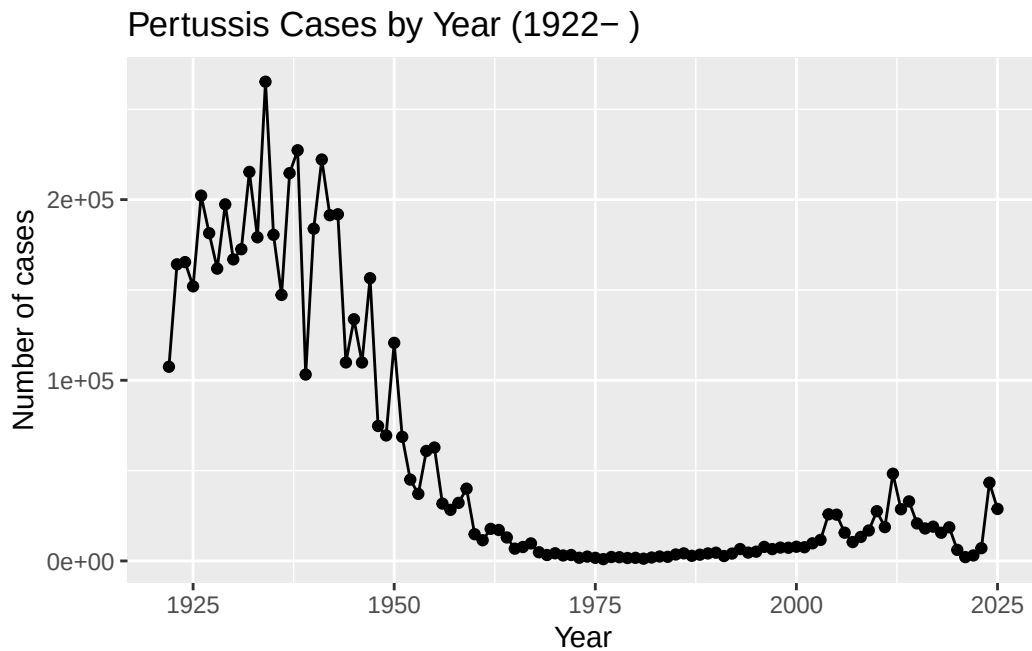
98	2019	18,617	Finalized
99	2020	6,124	Finalized
100	2021	2,116	Finalized
101	2022	3,044	Finalized
102	2023	7,063	Finalized
103	2024	43,321	Provisional
104	2025	28,783	Provisional

```
cdc$Number.of.Reported.Pertussis.Cases <- as.numeric(gsub(",", "", cdc$Number.of.Reported.Cases))
```

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
ggplot(cdc) +
  aes(Year, Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(x = "Year",
       y = "Number of cases",
       title = "Pertussis Cases by Year (1922- )")
```

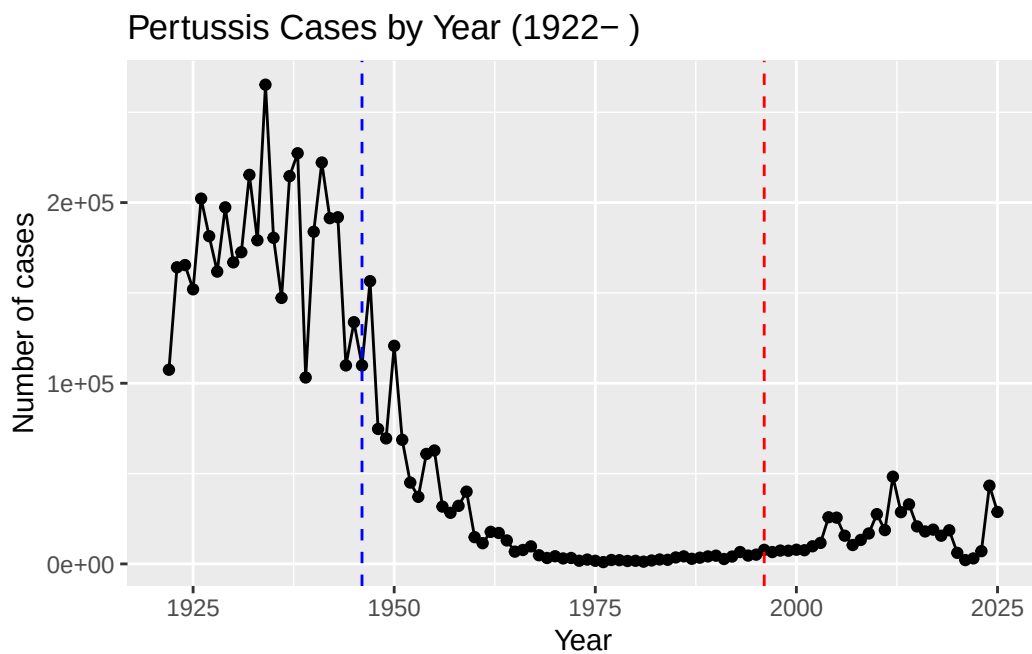


2. A tale of two vaccines (wP & aP)

Q2.

Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +  
  aes(Year, Number.of.Reported.Pertussis.Cases) +  
  geom_point() +  
  geom_line() +  
  geom_vline(xintercept=c(1946, 1996),  
            linetype = 2,  
            col = c("blue", "red")  
            ) +  
  labs(x = "Year", y = "Number of cases", title = "Pertussis Cases by Year (1922- )")
```



I notice that after the introduction of the wP vaccine (blue line), the number of observed cases started to decrease.

Q3.

Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine in 1996, the number of observed cases started increasing slowly. A possible explanation could be PCR-based testing turning more sensitive, vaccination hesitancy, bacterial evolution, or waning immunity in adolescents with the newer aP vaccine in comparison to the older wP vaccine.

3. Exploring CMI-PB data

The CMI-PB API returns JSON data

```
# Allows us to read, write and process JSON data
library(jsonlite)
```

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4.

How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

There are 87 aP and 85 wP infancy vaccinated subjects in the dataset.

Q5.

How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

Female	Male
112	60

There are 112 Female patients and 60 Male patients.

Q6.

What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

Breakdown of race and biological sex displayed in table above.

Side-Note: Working with dates**Q7.**

Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?


```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
# Use todays date to calculate age in days
subject$age <- today() - ymd(subject$year_of_birth)
```

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.5.2

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
#Split by aP group, summarize aP ages
ap <- subject %>% filter(infancy_vac == "aP")
round( summary( time_length( ap$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
# Split by wP group, summarize wP ages
wp <- subject %>% filter(infancy_vac == "wP")
round(summary(time_length( wp$age, "years" )))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

The average age of wP individuals is 37 and 28 for aP individuals. Based on the results, it does seem that they are significantly different as the wP values are greater, and the Median, Mean, 3rd Qu., and Max wP values different and much greater than the aP values.

Q8.

Determine the age of all individuals at time of boost?

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

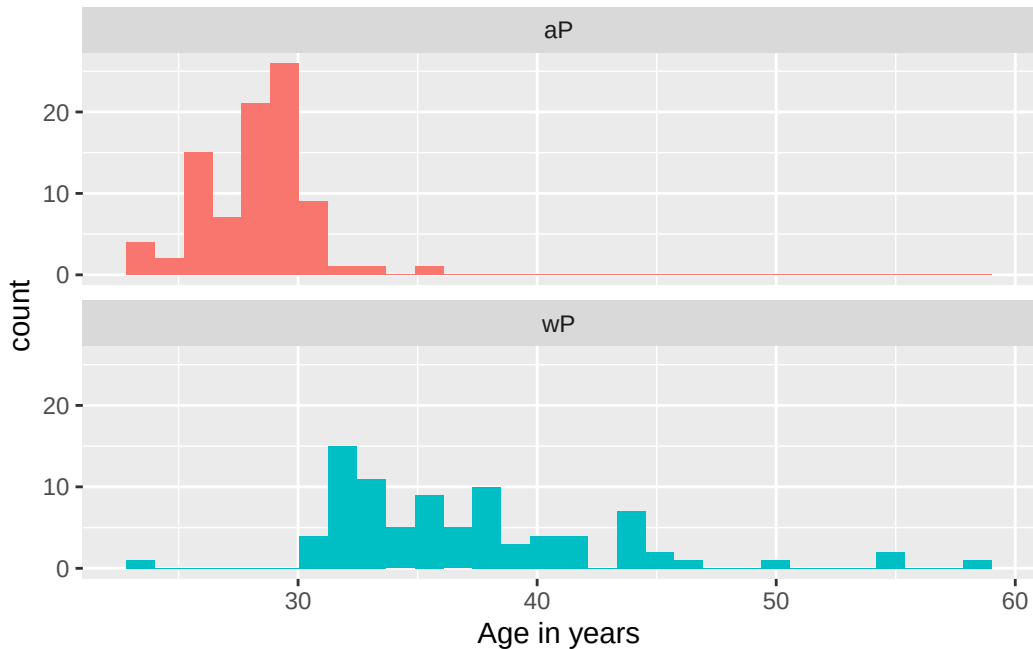
```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9.

With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
# Or use wilcox.test()
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

I think these two groups are significantly different as the p-value of 2.372101e-23 indicates that they are different. The aP group has more cases focused in those who are younger with ages ~25-30, while the wP group has more cases spread across all ages, mainly focused in the age range of 30-45, with some cases found in ages 45+.

Joining multiple tables

Read the specimen and ab_titer tables into R and store the data as specimen and titer named data frames.

```
# Complete the API URLs...
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9a.

Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject)
```

Joining with `by = join\_by(subject\_id)`

```
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

	specimen_id	subject_id	actual_day_relative_to_boost				
1	1	1	-3				
2	2	1	1				
3	3	1	3				
4	4	1	7				
5	5	1	11				
6	6	1	32				

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	1	Blood	2	wP	Female
3	3	Blood	3	wP	Female
4	7	Blood	4	wP	Female
5	14	Blood	5	wP	Female
6	30	Blood	6	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age
1	14759 days
2	14759 days
3	14759 days

```
4 14759 days
5 14759 days
6 14759 days
```

Q10.

Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
dim(abdata)
```

```
[1] 52576    21
```

Q11.

How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
 IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698 5389 10117 10124 10124 10124
```

We have **6698** specimens for IgE, **5389** for IgG, **10117** for IgG1 and **10124** for IgG2, IgG3, IgG4.

Q12.

What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520         8085         7301         5670
```

The different \$dataset values in abdata are 31520 (2020 dataset), 8085 (2021 dataset), 7301 (2022 dataset), and 5670 (2023 dataset). The number of rows for the most “recent” dataset, 5670 in the 2023 dataset, is much lower than the other datasets, especially in comparison to the number of rows in the 2020 dataset, 31520.

4. Examine IgG Ab titer levels

Now using our joined/merged/linked abdata dataset `filter()` for IgG isotype.

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female

3		0	Blood	1	wP	Female
4		0	Blood	1	wP	Female
5		0	Blood	1	wP	Female
6		0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

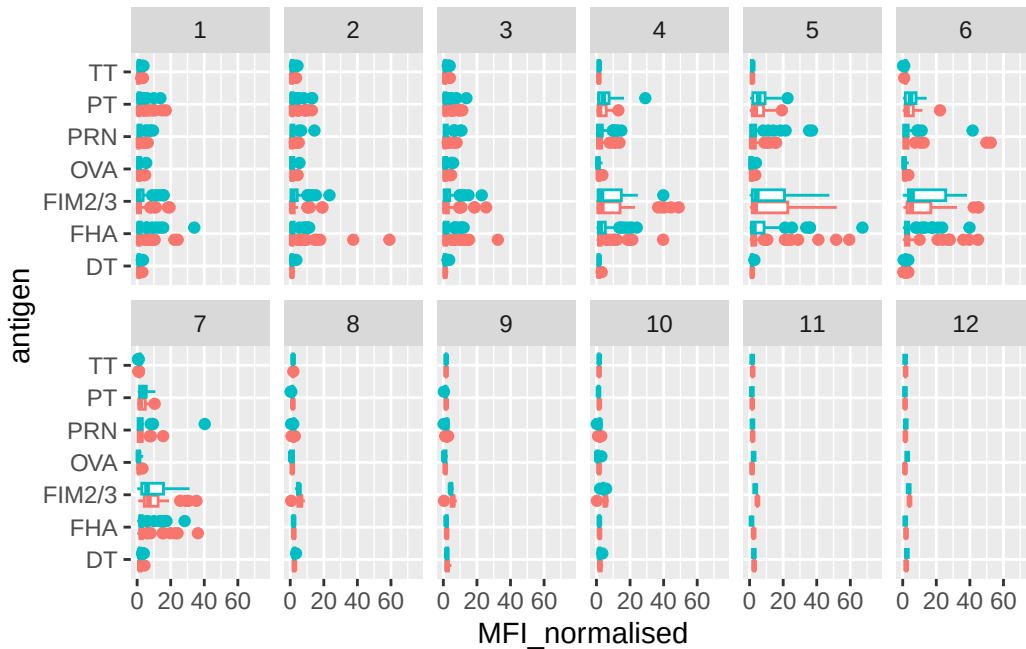
	age
1	14759 days
2	14759 days
3	14759 days
4	15855 days
5	15855 days
6	15855 days

Q13.

Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q14.

What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The FHA and FIM2/3 antigens show differences in the level of IgG antibody titers recognizing them, as well as little changes in PT and PRN, but not as much change in TT, DT, and OVA. This is because these antigens, FHA, FIM2/3, PT, and PRN, are Bordetella pertussis antigens, and they exhibit significantly elevated antibody titers, demonstrating an effective immune response directed at pertussis-specific antigens. OVA is used as a negative control antigen and thus not targeted by the vaccine and TT and DT are more stable comparison antigens. If the immune response is changing after pertussi exposure or vaccination, the antigens showing differences would most likely show it.

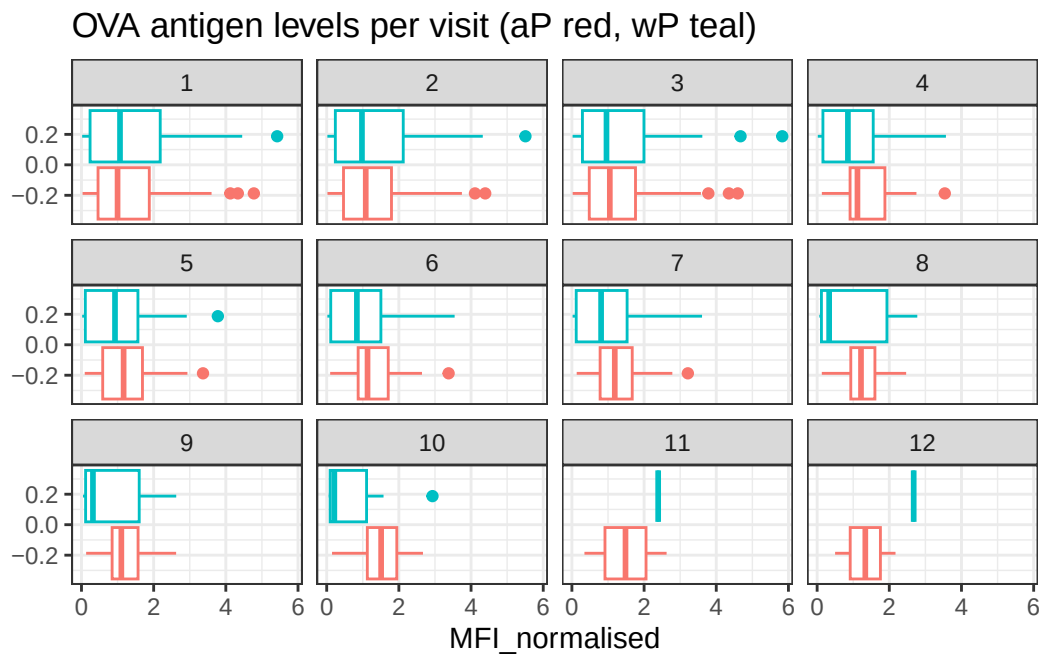
Q15.

Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).


```

filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()+
  labs(title = "OVA antigen levels per visit (aP red, wP teal)")

```



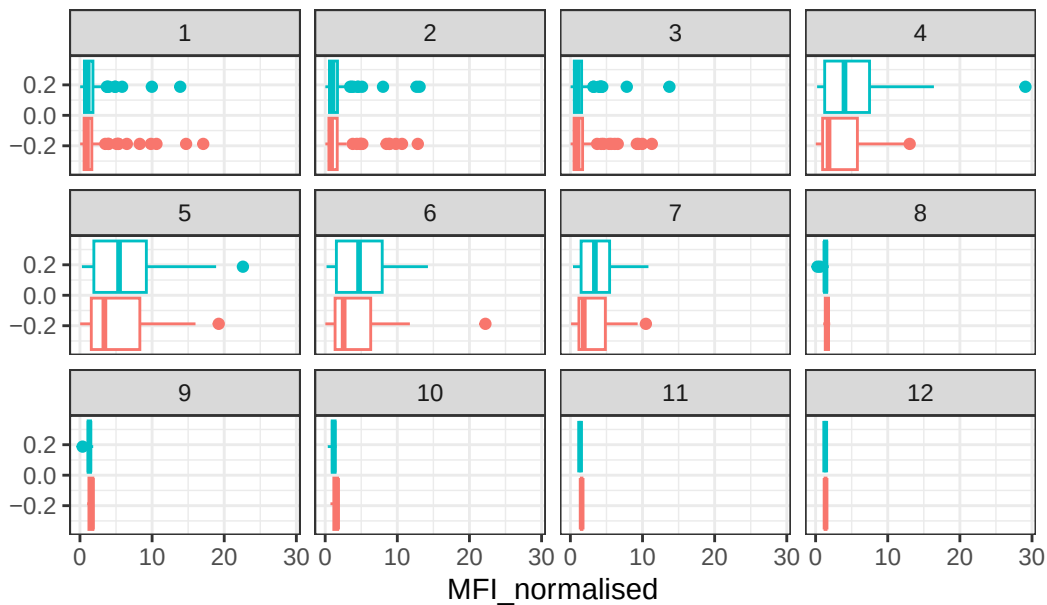
and the same for antigen=="PT"

```

filter(igg, antigen=="PT") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()+
  labs(title = "PT antigen levels per visit (aP red, wP teal)")

```

PT antigen levels per visit (aP red, wP teal)



Q16.

What do you notice about these two antigens time courses and the PT data in particular?

As time passes, OVA levels stay pretty consistently low among all visits as OVA is the negative control antigen. Over time, PT antibody levels are low overall during visits 1-3, start to increase in visits 4-7, and then decreases to baseline for visits 8-12, indicating that the PT response is variable. This comparison of OVA vs. PT shows that antigen-specific immunity grows against the vaccine.

Q17.

Do you see any clear difference in aP vs. wP responses?

There are not many clear differences in aP vs. wP responses, they seem pretty similar. Both groups increase in the PT antibody titers, but eventually decrease. Thus, both vaccine types can result in immune responses.

Lets finish this section by looking at the 2021 dataset IgG PT antigen levels time-course:

```

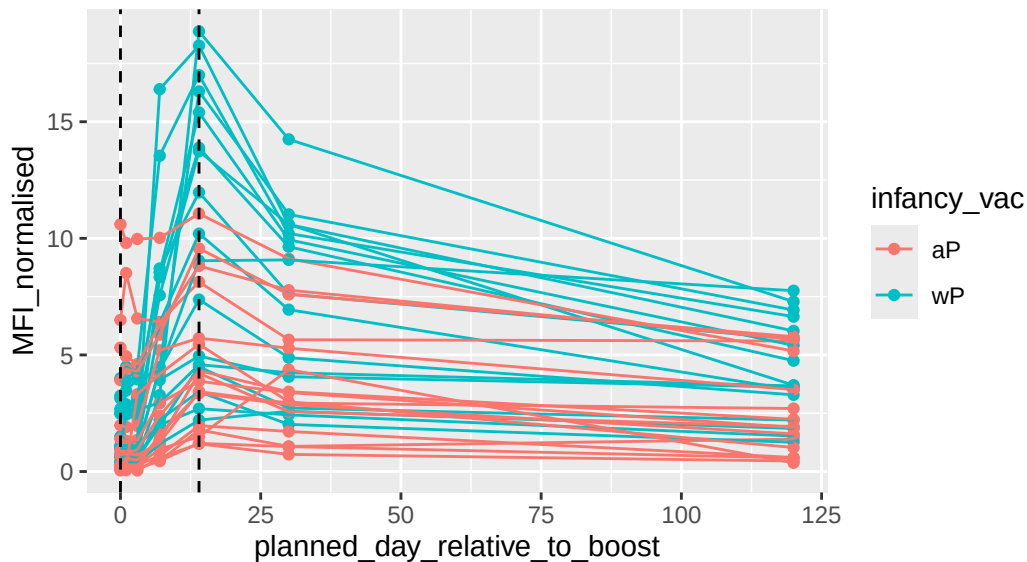
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")

```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



```

abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

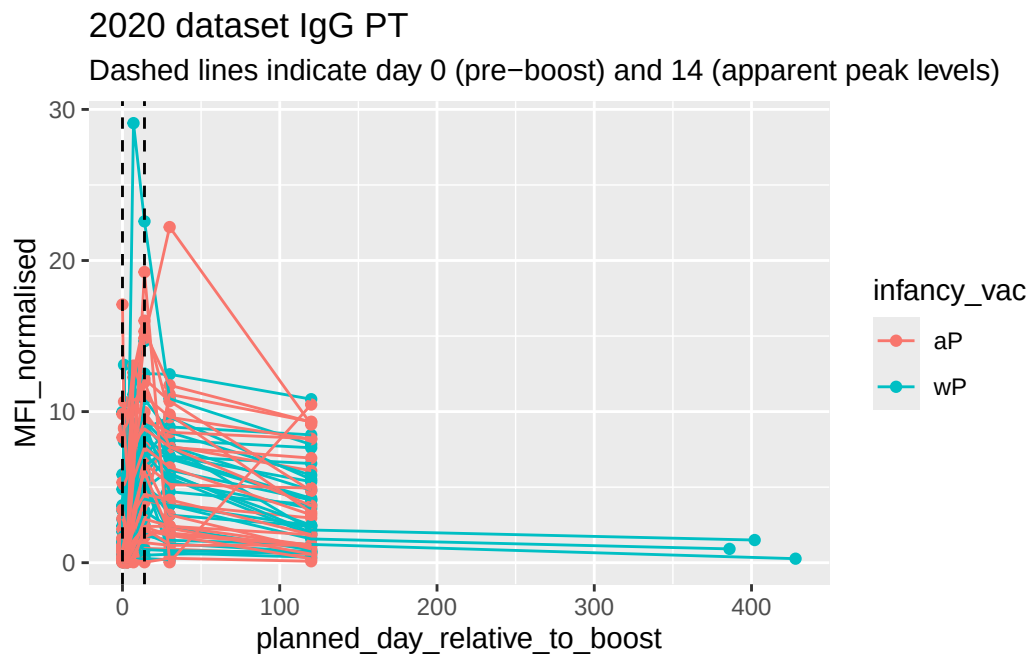
abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,

```

```

y=MFI_normalised,
col=infancy_vac,
group=subject_id) +
geom_point() +
geom_line() +
geom_vline(xintercept=0, linetype="dashed") +
geom_vline(xintercept=14, linetype="dashed") +
labs(title="2020 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")

```



Q18.

Does this trend look similar for the 2020 dataset?

Yes, this trend looks pretty similar for the 2020 dataset. The IgG PT levels are peak around day 14 before their decline.

5. Obtaining CMI-PB RNASeq data

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENS00000211896.7"

rna <- read_json(url, simplifyVector = TRUE)

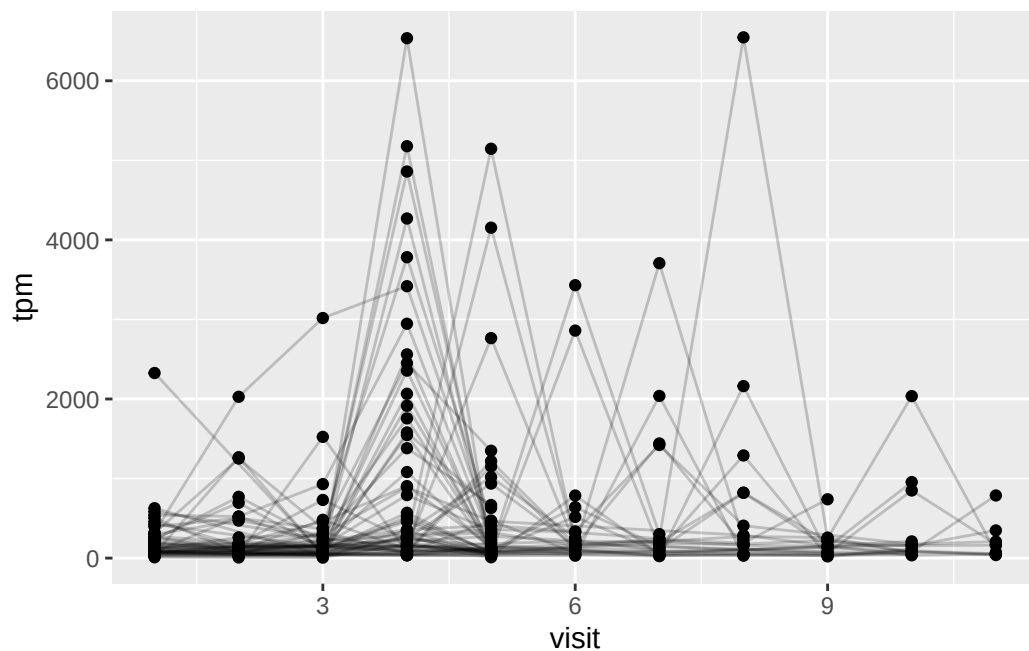
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19.

Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



Q20.:

What do you notice about the expression of this gene (i.e. when is it at its maximum level)?

I notice that the expression of this gene is at its maximum level right after booster vaccination. The IGHG1 expression is mostly low at many visits, but then spikes early with a maximum level at visit 4 and then randomly later on at visit 8.

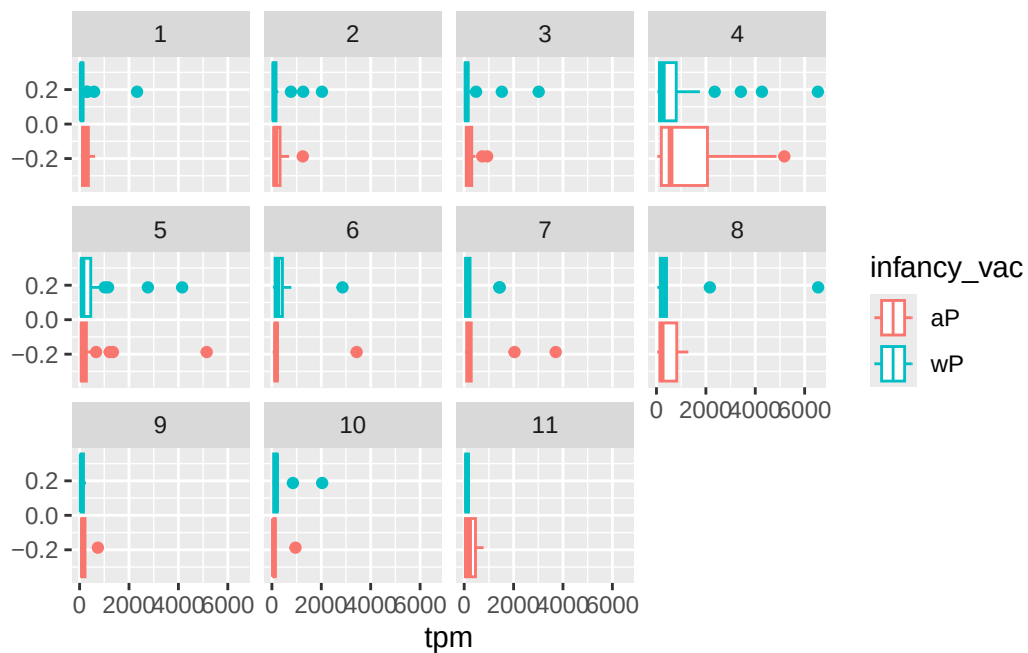
Q21.

Does this pattern in time match the trend of antibody titer data? If not, why not?

No, this pattern in time does not match the trend of antibody titer data because the IGHG1 expression level peaks at visit 4, while the antibody titer data displays variable data and peaks later on. IGHG1 expression measures mRNA inside cells and antibody titer measures secreted antibody protein in blood. IGHG1 expression levels peak earlier and antibody titers peak later because the mRNA must first be translated into proteins before those antibodies are secreted into the bloodstream.

We can dig deeper and color and/or facet by infancy_vac status:

```
ggplot(ssrna) +  
  aes(tpm, col=infancy_vac) +  
  geom_boxplot() +  
  facet_wrap(vars(visit))
```



There is however no obvious wP vs. aP differences here even if we focus in on a particular visit:

```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```

